

Problem

Meter readings for a three-phase wye-connected alternator supplying power to a motor indicate that the line voltages are 330 V, the line currents are 8.4 A, and the total line power is 4.5 kW. Find:

(a) the load in VA

(b) the load pf

(c) the phase current

(d) the phase voltage

Step-by-step solution

Step 1 of 4

$$\begin{aligned} \text{(A) } S &= \sqrt{3} V_L I_L \\ &= \sqrt{3} (330)(8.4) \\ &= 4801 \text{ VA} . \end{aligned}$$

Step 2 of 4

$$\begin{aligned} \text{(B) } P &= S \cos \theta \Rightarrow P f = \cos \theta = \frac{P}{S} \\ P f &= \frac{4500}{4801.24} = 0.9372 . \end{aligned}$$

Step 3 of 4

(C) For a WYE – connected load,

$$I_p = I_L = 8.4 \text{ A} .$$

Step 4 of 4

$$\text{(D) } V_p = \frac{V_L}{\sqrt{3}} = \frac{330}{\sqrt{3}} = 190.33 \text{ V} .$$